

MATS APPLICATION: TRANSIENT ESCAPE FROM A REFUSAL TRAJECTORY IN AN INSTRUCTION-TUNED TRANSFORMER

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Part 1. 3-Page Executive Summary

EXECUTIVE SUMMARY

Research Question. I investigated whether jailbreak outcomes in an instruction-tuned language model are associated with a localized change in internal representation dynamics. In particular, I asked whether prompts that lead to non-refusal or provisionally successful jailbreak outcomes exhibit a measurable departure from an internal refusal-associated trajectory at particular transformer layers.

I chose this question because numerous mechanistic interpretability analyses treat model activations as static objects, where one extracts a hidden state at a layer and asks whether it linearly separates two behaviors. Although useful, such analyses can confound behavior with prompt topic, wording, or style. I therefore tested a more dynamical hypothesis by asking whether the path of the residual-stream representation across layers contains a localized signature of a behavioral transition.

Initial Approach and Why I Changed It. I first tested a simple hidden-state classification setup using benign prompts and safety-related prompts. For Qwen2.5-3B-Instruct, I extracted the residual-stream representation of the final prompt token at each layer and constructed layerwise directions separating the two classes. The result initially appeared extremely strong with near-perfect AUROC across most layers. However, this result was not sufficient to support a mechanistic interpretation. The benign and safety prompts differed substantially in semantic content and lexical cues, so the classifier could have been detecting concepts such as authorization, credentials, privacy, or security rather than anything specifically related to refusal behavior. Therefore, I constructed a synthetic set on the same topic containing benign, unsafe, and contextual-safety variants of the same underlying topic. Topic-held-out classification remained very strong. A further comparison between direct unsafe prompts and contextual safety prompts was also nearly perfectly separable. This indicated that prompt framing itself was strongly encoded and that static classification remained too easy to interpret mechanistically. This negative result changed the direction of the project. Instead of asking whether behaviors occupy different regions of the activation space, I asked whether the residual-stream trajectory undergoes a localized departure from a refusal-associated path.

Dynamical Formulation. For each prompt x , I extracted the final-prompt-token residual representation across the network, $h_0(x), h_1(x), \dots, h_L(x)$, where $h_\ell(x)$ denotes the hidden representation at layer ℓ . Because the residual-stream norm grows substantially with depth, I normalized each state,

$$\hat{h}_\ell(x) = \frac{h_\ell(x)}{\|h_\ell(x)\|_2 + \varepsilon}.$$

I then constructed an independent refusal trajectory from a separate collection of matched unsafe prompts that the model consistently refused. Let r_ℓ denote the normalized refusal centroid at layer ℓ . For a new prompt x , define

its distance from the refusal trajectory by

$$d_\ell(x) = \left\| \hat{h}_\ell(x) - r_\ell \right\|_2.$$

Then I measured the layerwise change in this distance using the following formula:

$$e_\ell(x) = d_{\ell+1}(x) - d_\ell(x).$$

Positive values of $e_\ell(x)$ indicate that the representation is moving away from the refusal trajectory at the transition from layer ℓ to $\ell + 1$. Exploratory analysis showed the most visible group separation in a late-layer region, approximately layers 22–30. Therefore, I define the escape score as follows:

$$E(x) := \max_{\ell=22, \dots, 30} e_\ell(x).$$

Model, Data, and Experimental Setup. I used Qwen2.5-3B-Instruct and sampled 60 adversarial prompts from the JailbreakBench judge-comparison dataset. All model inference and hidden-state extraction were performed locally using PyTorch-MPS – M5. For each prompt, I saved the generated response and the final-prompt-token residual representation across 37 hidden-state levels, each of dimension 2048. The behavioral outcome labels used for the final quantitative analysis were provisional. I combined an explicit-refusal detector with a semantic goal-response similarity measure and assigned each output to one of three categories, given as failed attack, successful attack, or ambiguous. Three ambiguous cases were excluded, leaving 57 examples - 22 provisional failures, 35 provisional successes.

Main Result. Overall, layer-to-layer representation movement was very similar between explicit refusals and non-refusals. Thus, the main effect was not simply that one class exhibited larger hidden-state movement throughout the network. Instead, the more distinctive behavior appeared in the distance from the independent refusal trajectory. Provisionally successful cases showed a larger localized late-layer departure, particularly around layers 22–30. As shown in Figure 1, the two outcome groups exhibit broadly similar layer-to-layer dynamics across much of the network, but the provisionally successful cases show a larger localized departure from the independent refusal trajectory in the late-layer region, particularly across layers 22–30.

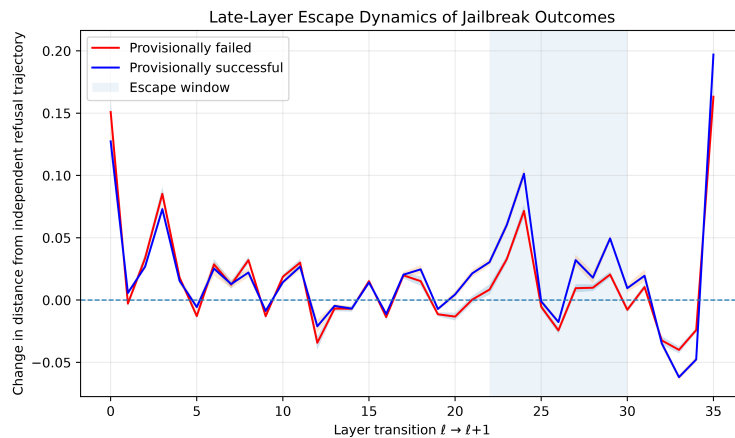


FIGURE 1. **Late-layer escape from an independently defined refusal trajectory.** Successful attacks exhibit a larger localized departure from the refusal trajectory in the late-layer region used to define the escape score, while overall layer-to-layer dynamics remain broadly similar across groups.

The mean escape scores were

$$\overline{E}_{\text{failure}} = 0.0715, \quad \overline{E}_{\text{success}} = 0.1066.$$

Using $E(x)$ alone to predict the provisional outcome,

$$\text{AUROC} = 0.8325.$$

A 5000-trial label-permutation test gave $p \approx 2.0 \times 10^{-4}$, and the bootstrap 95% confidence interval for AUROC was [0.693, 0.943]. I **interpret** this as preliminary evidence that jailbreak-related behavior may be associated with a localized late-layer departure from a refusal-associated internal trajectory rather than with a globally larger representational motion.

Strongest Evidence Against the Hypothesis. The strongest counterevidence came from the initial static classification experiments. They produced a nearly perfect separation, but follow-up controls showed that this result was strongly confounded by the semantic topic and prompt framing. This made it clear that high classification accuracy alone did not constitute mechanistic evidence. The second caution is that the final layer of the network exhibits a large representational jump for both successful and failed groups. Therefore, I do not interpret this final transition as jailbreak-specific. A third limitation is that the final behavioral labels are heuristic rather than produced by an independent established jailbreak judge. I attempted to use the JailbreakBench judge implementation, but package and environment compatibility prevented a clean evaluation within the available research window. Therefore, I consider the reported AUROC as exploratory rather than definitive.

Limitations and Next Steps. The present study uses a single model, a small sample of 60 benchmark prompts, and a layer window that emerged during the exploratory analysis rather than being preregistered. The behavioral labels are provisional, and the analysis is correlational. I have not demonstrated that manipulating the identified late-layer region causally changes the refusal behavior. The most important next experiment is to reevaluate the same model responses using an independent jailbreak judge and to replicate the escape analysis on a second modern instruction-tuned model. If the late-layer effect persists, a causal intervention, such as activation steering or activation patching, could test whether suppressing or amplifying the identified escape changes the refusal behavior. The main lesson from this project is therefore not merely that a hidden-state statistic can predict jailbreak behavior. The more interesting observation is that static separation was too easy to trust, while a trajectory-based analysis produced a narrower and more falsifiable hypothesis about where refusal behavior may begin to diverge internally.

Part II: Main Project Write-Up

MAIN PROJECT

The purpose of this study was to investigate whether refusal-related behavior in an instruction-tuned language model admits a useful internal dynamical description. The central object of analysis is not a single hidden state in isolation, but the sequence of residual-stream representations generated as a prompt propagates through successive transformer layers. For an input prompt x , let $h_0(x), h_1(x), \dots, h_L(x)$ denote the final-prompt-token residual representations across the network. The initial question was whether safety-related behavioral outcomes could be distinguished from these internal states. During the course of the study, this question was substantially refined after several static classification tests proved to be too easy to interpret mechanistically. The final analysis therefore focuses on whether a prompt moves away from an independently constructed refusal-associated trajectory within a localized region of the network.

Model and Computational Setup. All experiments were performed with Qwen/Qwen2.5 – 3B – Instruct. The model was evaluated locally using PyTorch and Hugging Face Transformers on Apple Silicon through the MPS backend. For every prompt, I applied the model’s instruction chat template and extracted the hidden states from a forward pass before generation. From each hidden-state tensor, I retained the representation of the final prompt token. This produced one vector per layer, $h_\ell(x) \in \mathbb{R}^{2048}$, across 37 hidden-state levels. The resulting representation tensor for a collection of N prompts has shape $N \times 37 \times 2048$.

Generated responses were obtained using deterministic decoding (`do_sample = False`). This choice reduced output variability during the exploratory analysis and made repeated comparisons easier to interpret. A strong increase in residual-stream norm was observed with depth. For example, in an early pipeline check the norm increased from approximately 1.26 at the embedding level to approximately 186.6 at the final hidden-state level. Consequently, all trajectory analyses were carried out using normalized states $\hat{h}_\ell(x) = \frac{h_\ell(x)}{\|h_\ell(x)\|_2 + \varepsilon}$, with a small numerical stabilizer $\varepsilon > 0$.

Static Representation Experiments.

Initial Safety-versus-Benign Test. The first experiment used 60 synthetic prompts divided into three groups:

- (1) benign prompts,
- (2) direct safety-related prompts,
- (3) contextual safety-evaluation prompts.

Each group contained 20 examples. The direct safety prompts were consistently refused by the model, while benign prompts received ordinary helpful responses. First, I compared the layerwise centroids of the three groups. Raw Euclidean distances increased sharply in late layers, but this behavior was partially explained by the increasing residual-stream norm. After normalizing the representations, the safety-related groups remained substantially closer to each other than either was to the benign group.

I then constructed a layerwise direction from the difference between the benign and direct-safety centroids and evaluated its ability to separate the two classes. The resulting AUROC was essentially perfect across most layers. This result was not retained as evidence for a refusal-specific mechanism. The two groups differed substantially in semantic content and lexical cues, including concepts associated with authorization, security, privacy, and credentials. The experiment therefore did not isolate behavior from the topic.

Matched-Topic Control. To reduce this confounding, I constructed 30 matched prompt triplets, resulting in 90 prompts in total. Each triplet contained

- (1) a benign request,
- (2) an unsafe request,
- (3) a contextual safety-evaluation version,

all referring to the same underlying topic. For example, a single topic could concern password security, while varying only whether the user requested defensive guidance, unauthorized access, or a meta-level safety response. The activation tensor for this experiment had a shape $90 \times 37 \times 2048$.

To avoid topic leakage, I evaluated benign-versus-unsafe classification using topic-held-out splits, that is, all members of a given topic were assigned entirely to either the training or test set. Even under this stronger split, the simple representation direction remained highly predictive, reaching near-perfect AUROC in later layers. A second matched comparison between direct unsafe prompts and contextual safety prompts was also nearly perfectly separable. This showed that prompt framing itself was strongly represented and could dominate static classification performance. These experiments motivated the abandonment of static separability as the primary objective of the study.

Transition to a Trajectory-Based Analysis. The next question was whether refusal behavior could instead be characterized through the *evolution* of the residual stream across layers. For consecutive normalized representations, define

$$\Delta_\ell(x) = \hat{h}_{\ell+1}(x) - \hat{h}_\ell(x).$$

I first measured the layer-to-layer movement

$$g_\ell(x) = \|\Delta_\ell(x)\|_2.$$

When averaged across explicit refusals and responses without an explicit refusal phrase, the resulting curves were very similar throughout the network. This provided an important negative result, i.e. the behavioral distinction was not explained simply by one class having a uniformly larger residual-stream motion. This observation suggested measuring movement relative to a behaviorally meaningful reference rather than measuring total movement alone.

Independent Refusal-Trajectory Construction. A refusal reference was constructed from the independent matched topic dataset described above. Only the 30 `unsafe_refused` examples were used. Let $\mathcal{R}$ denote this independent set of consistently refused prompts. At each layer, define the refusal centroid

$$\tilde{r}_\ell = \frac{1}{|\mathcal{R}|} \sum_{x \in \mathcal{R}} \hat{h}_\ell(x).$$

Because the average of unit vectors is not necessarily itself unit norm, the reference trajectory was renormalized as in $r_\ell = \frac{\tilde{r}_\ell}{\|\tilde{r}_\ell\|_2 + \varepsilon}$. This construction is intentionally external to the JailbreakBench examples used in the final analysis. The same examples whose escape scores were measured were therefore not used to define the refusal centroid. For a new prompt x , the layerwise distance from the refusal trajectory is $d_\ell(x) = \|\hat{h}_\ell(x) - r_\ell\|_2$. The local escape increment is then $e_\ell(x) = d_{\ell+1}(x) - d_\ell(x)$. Thus, $e_\ell(x) > 0$ means that the representation moves farther away from the refusal-associated trajectory between layers ℓ and $\ell + 1$, while $e_\ell(x) < 0$ means that it moves closer.

JailbreakBench Experiment.

Prompt Set. For the final experiment, I sampled 60 adversarial prompts from the JailbreakBench judge-comparison dataset. Each prompt was passed through Qwen2.5-3B-Instruct using the same activation-extraction procedure described above. The resulting activation tensor had a shape $60 \times 37 \times 2048$. For each example, I stored

- the benchmark goal,
- the adversarial prompt,

- the generated Qwen response,
- the layerwise hidden states,
- the resulting escape trajectory.

Provisional Behavioral Labels. The available JailbreakBench human labels apply to responses stored in the benchmark and therefore cannot be directly assigned to newly generated Qwen responses. In the following, I used a provisional automated labeling procedure based on two signals:

- (1) explicit refusal-language detection;
- (2) semantic similarity between the benchmark goal and the generated response.

The responses were assigned to one of three categories given as follows:

$$0 = \text{provisional failure}, \quad 1 = \text{provisional success}, \quad 2 = \text{ambiguous}.$$

Three ambiguous cases were excluded from the quantitative analysis. The final sample therefore contained $N = 57$, with

$$22 \text{ provisional failures} \quad \text{and} \quad 35 \text{ provisional successes}.$$

These labels are not treated as definitive ground truths. Their provisional status is a central limitation of the present study.

Escape-Score Definition. The layerwise escape curves showed a visible group difference in a late-layer region, while the final network transition produced a large jump for both groups. To avoid incorporating the final shared jump into the metric, I used the exploratory window $\ell = 22, \dots, 30$. The scalar escape score is $E(x) = \max_{\ell=22, \dots, 30} e_{\ell}(x)$. Thus, $E(x)$ records the largest positive movement away from the independent refusal trajectory within the selected late-layer region. The window was selected during exploratory analysis rather than specified before observing the data. Therefore, I treat the resulting statistic as an exploratory candidate marker, not as a preregistered test.

Quantitative Results. The mean escape scores differed substantially between the two provisional outcome groups: $\bar{E}_{\text{failure}} = 0.0714727$, and $\bar{E}_{\text{success}} = 0.10659051$. Using $E(x)$ as a scalar predictor of the provisional outcome yields AUROC = 0.8324675. A label-permutation test with 5000 random permutations produced $p = 0.00019996$. The bootstrap 95% confidence interval for the AUROC was [0.6932, 0.9431]. These results indicate that the escape score contains substantial information about the provisional behavioral outcome in this sample. Importantly, the corresponding group difference is localized rather than global. The overall norm of the layer-to-layer residual-stream change was similar across groups, whereas the difference became more visible when motion was measured relative to the independent refusal reference.

Sanity Checks and Alternative Explanations. Several checks materially changed the interpretation of the study.

Residual-Stream Scale. The raw Euclidean centroid distances increased substantially with the depth of the network. Because hidden-state norms also increased strongly, raw distance alone was not interpretable. Repeating the analysis after unit normalization preserved the qualitative group structure, showing that the observed geometry was not solely an artifact of residual-stream scale.

Static Classification Confounds. Near-perfect static classification initially appeared promising, but matched topic and framing controls showed that the same type of separability persisted when lexical framing alone changed. Static AUROC was therefore not sufficient evidence for a refusal-specific internal mechanism.

Independent Reference Set. An early version of the trajectory analysis defined the refusal centroid from the same examples subsequently measured against it. This could artificially reduce the distance of the refusal group. Reconstructing the refusal trajectory from the independent matched dataset preserved the late-layer escape pattern.

Overall Representation Motion. The magnitude $\|\hat{h}_{\ell+1}(x) - \hat{h}_{\ell}(x)\|_2$ was broadly similar between the two groups. Therefore, the observed effect is not well described as a uniformly increased internal activity.

Shared Final-Layer Jump. Both outcome groups exhibited a large final transition. Since this effect was shared, it was excluded from the interpretation of the candidate jailbreak-associated region.

What the Current Result Does and Does Not Establish. The present experiment supports a narrow empirical statement that is provided as follows:

Within this exploratory Qwen2.5-3B-Instruct experiment, prompts assigned a provisionally successful outcome tend to exhibit a larger localized late-layer departure from an independently constructed refusal trajectory than prompts assigned a provisional failure.

The analysis does *not* establish that escape is the causal mechanism of jailbreak behavior. It also does not establish that the same layer region or direction generalizes across models. The measured association could reflect a downstream consequence of another internal computation, rather than the computation responsible for the behavioral transition itself.

Limitations. The main limitations are as follows.

- (1) **Outcome labeling.** The final labels are heuristic and provisional, rather than produced by an independent established jailbreak judge.
- (2) **Single model.** The analysis was performed only on Qwen2.5-3B-Instruct.
- (3) **Sample size.** The quantitative analysis contains only 57 non-ambiguous examples.
- (4) **Exploratory window selection.** Layers 22–30 were selected after inspecting the trajectory curves. The reported AUROC should therefore be interpreted as exploratory.
- (5) **No causal intervention.** No activation patching, steering, or other intervention was performed to test whether changing the identified trajectory changes the behavior of the model.
- (6) **Single-token representation.** The present analysis uses the final prompt token. It does not characterize how the same phenomenon evolves across all input tokens or during autoregressive generation.

Next Experiments. The most important next step is to replace the provisional response labels with an independent jailbreak evaluator and rerun the frozen escape-score analysis without changing the layer window or metric. A second test would replicate the same protocol on another modern instruction-tuned model. A cross-model result would help to distinguish a model-specific geometric effect from a more general feature of refusal processing. If the late-layer escape remains reproducible, then the next question becomes causal. One possible experiment is activation steering along the local direction away from the refusal trajectory. Another is activation patching between successful and failed examples in the identified late-layer region. Such interventions would test whether the observed escape merely predicts behavior or participates directly in producing it.

Research Takeaway. The main methodological lesson from the study is that high predictive accuracy in the activation space can be misleading when prompt content and framing are not controlled. The initial static results looked numerically stronger than the final trajectory result, but they supported a substantially weaker mechanistic interpretation. The trajectory formulation produced a more constrained empirical object, that is, a localized change in distance from an independent refusal-associated reference. Its current predictive performance is encouraging, but

its value lies primarily in producing a concrete hypothesis that can be independently judged, replicated, and causally tested.

Code and Reproducibility. The code, figures and aggregate results for the project are available at the project site.

`https://github.com/himanshuvnm/transient-refusal-dynamics`

The repository contains the model-analysis workflow, representation extraction, trajectory construction, escape-score computation, and statistical evaluation. Raw harmful prompt-response dumps are not included.

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